Homework 6

Due Tuesday, October 20, 2026 by 11:59 PM on Gradescope

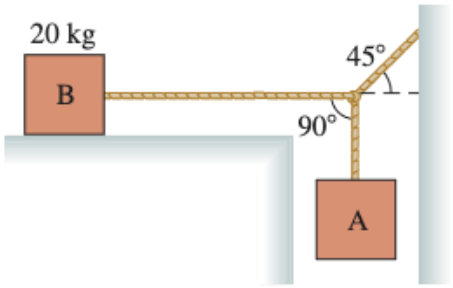
Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

This homework contains problems related forces, Newton's second and third laws, collections of objects, acceleration constraints, massless ropes, and massless, frictionless pulleys.

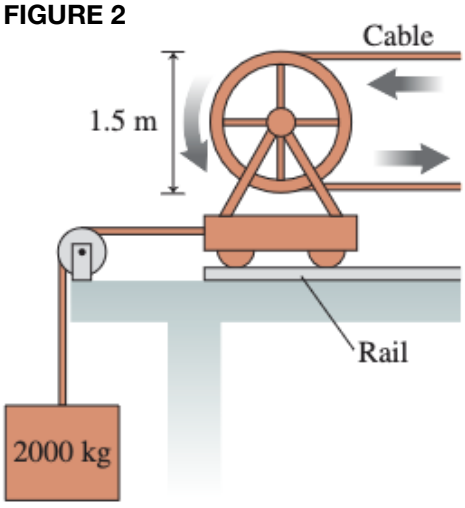
FIGURE 1



1. Block B in Figure 1 rests on a surface for which the static and kinetic coefficients of friction are 0.60 and 0.40, respectively. The ropes are massless. What is the maximum mass of block A for which the system remains in equilibrium?
2.
   a. How much force does an $80\,\mathrm{kg}$ astronaut exert on his chair while sitting at rest on the launch pad?
   b. How much force does the astronaut exert on his chair while accelerating straight up at $10\,\mathrm{m/s^2}$?
3. The foot of a $55\,\mathrm{kg}$ sprinter is on the ground for $0.25\,\mathrm{s}$ while her body accelerates from rest to $2.0\,\mathrm{m/s}$.
   a. Is the friction between her foot and the ground static friction or kinetic friction?
   b. What is the magnitude of the friction force?Your forehead can withstand a force of about

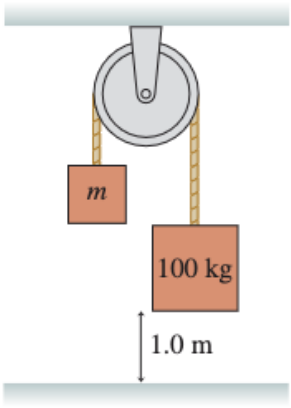
4. Your forehead can withstand a force of about $6\,\mathrm{kN}$ before fracturing, while your cheekbone can withstand only about $1.3\,\mathrm{kN}$. Suppose a $140\,\mathrm{g}$ baseball traveling at $30\,\mathrm{m/s}$ strikes your head and stops in a $1.5\,\mathrm{ms}$.
   a. What is the magnitude of the force that stops the baseball?
   b. What force does the ball exert on your head? Explain.
   c. Are you in danger of fracturing if the ball hits you in the forehead? On the cheek?Blocks with masses of $1\,\mathrm{kg}$, $2\,\mathrm{kg}$, and $3\,\mathrm{kg}$ are lined up in a row on a frictionless table. All three are pushed forward by a $12\,\mathrm{N}$ force applied to the $1\,\mathrm{kg}$ block.
5. Blocks with masses of $1\,\mathrm{kg}$, $2\,\mathrm{kg}$, and $3\,\mathrm{kg}$ and are lined up in a row on a frictionless table. All three are pushed forward by a $12\,\mathrm{N}$ force applied to the $1\,\mathrm{kg}$ block.
   a. How much force does the $2\,\mathrm{kg}$ block exert on the $3\,\mathrm{kg}$ block?
   b. How much force does the $2\,\mathrm{kg}$ block exert on the $1\,\mathrm{kg}$ block?
6. A $2.0\,\mathrm{m}$ long, $500\,\mathrm{g}$ rope pulled a $10\,\mathrm{kg}$ block of ice across a horizontal, frictionless surface. The block accelerates at $2.0\,\mathrm{m/s^2}$. How much force pulls forward on (a) the ice, (b) the rope? Assume that the rope is perfectly horizontal.

FIGURE 2



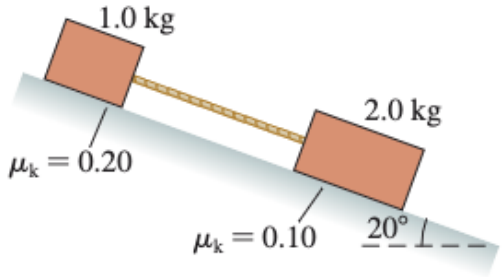
7. The cable cars in San Francisco are pulled along their tracks by an underground steel cable that moves along at $9.5\ \text{mph}$. The cable is driven by large motors at a central power station and extends, via an intricate pulley arrangement, for several miles beneath the city streets. The length of a cable stretches by up to $100\ \text{ft}$ during its lifetime. To keep the tension constant, the cable passes around a $1.5\ \text{m}$ diameter "tension pulley" that rolls back and forth on rails, as shown in Figure 2. A $2000\ \text{kg}$ block is attached to the tensioning pulley's cart, via a rope and pulley, and is suspended in a deep hole. What is the tension in the cable car's cable?

FIGURE 3



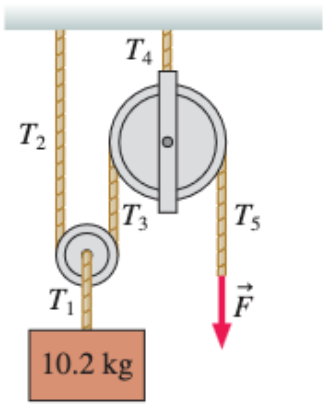
8. The $100\,\text{kg}$ block in Figure 3 takes $6.0\,\text{s}$ to reach the floor after being released from rest. What is the mass of the block on the left? The pulley is massless and frictionless.

FIGURE 4



9. The two blocks in Figure 4 are sliding down the incline. What's the tension in the massless string?

FIGURE 5



10. The $10.2\,\text{kg}$ block in figure 5 is held in place by a force applied to a rope passing over two massless, frictionless pulleys. Find the tensions T_1 to T_5 and the magnitude of force $\vec{F}$.

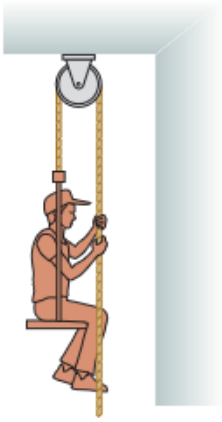
11. A $4.0\,\text{kg}$ box is on a frictionless $35°$ slope and is connected via a massless string over a massless, frictionless pulley to a hanging $2.0\,\text{kg}$ weight.

    a. What is the tension in the string if the $4.0\,\text{kg}$ box is held in place, so that it cannot move?

    b. If the box is then released, which way will it move on the slope?

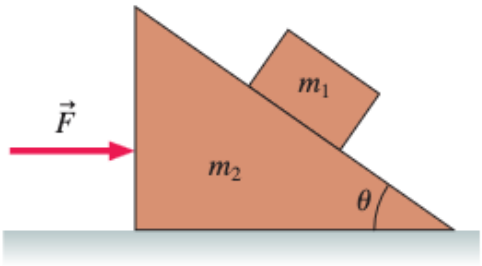
    c. What is the tension in the string once the box begins to move?

FIGURE 6



12. A house painter uses the chair-and-pulley arrangement of Figure 6 to lift himself up the side of a house. The painter's mass is $70\,\mathrm{kg}$ and the chair's mass is $10\,\mathrm{kg}$. With what force must he pull down on the rope in order to accelerate upward at $0.20\,\mathrm{m/s^2}$?

FIGURE 7



13. Find an expression for the magnitude of the horizontal force $\vec{F}$ in Figure 7 for which m_1 does not slip either up or down along the wedge. All surfaces are frictionless.